

Unit 5, Lesson 3: Solving Problems Involving Perimeter and Area of Geometric Figures

Benchmark

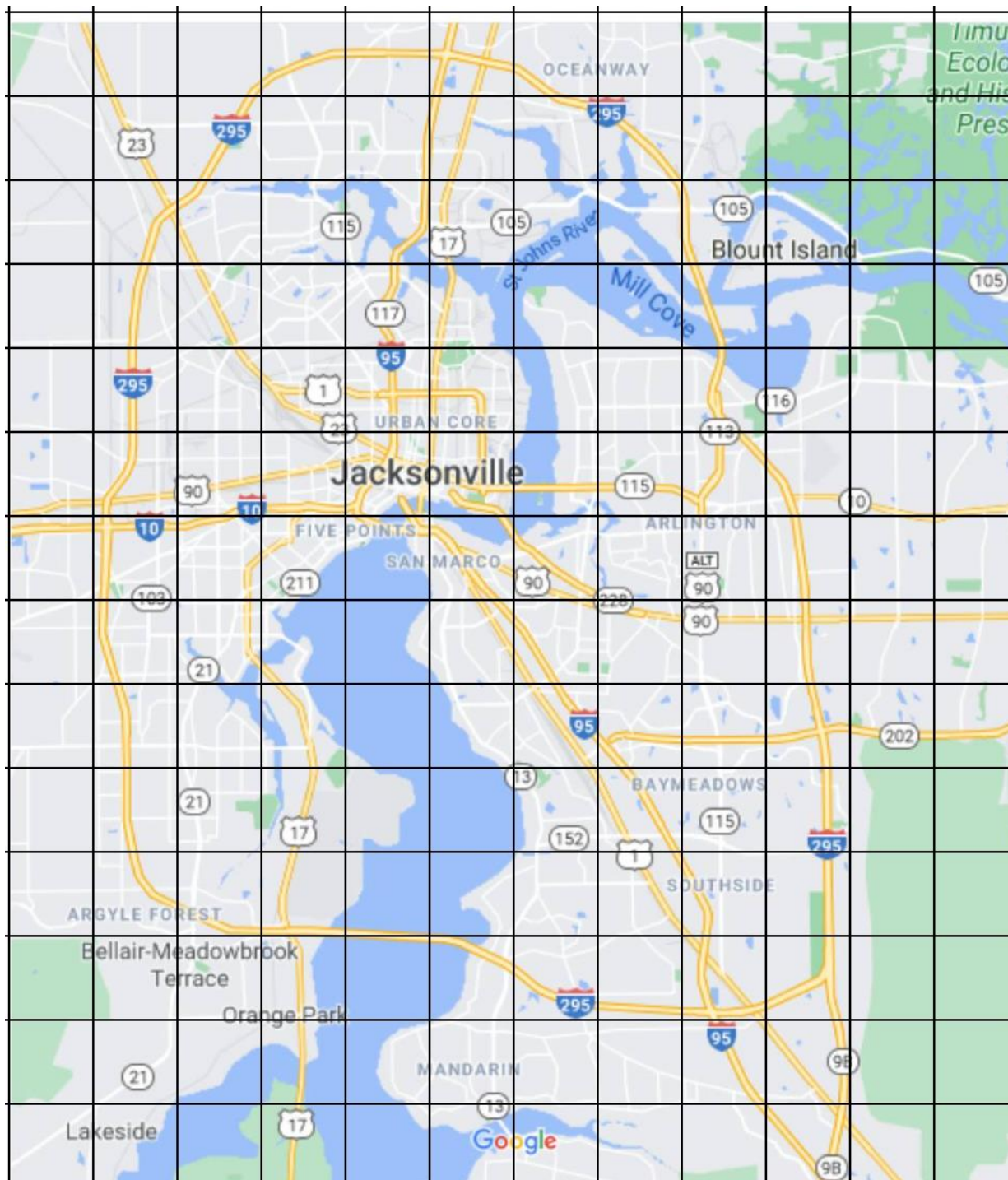
MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Learning Target

- ☐ I can solve mathematical and real-world problems involving dimensions, perimeters, and areas of geometric figures.

5.3.1 Warm-Up: Notice and Wonder

A map of Jacksonville and its surrounding area is shown.



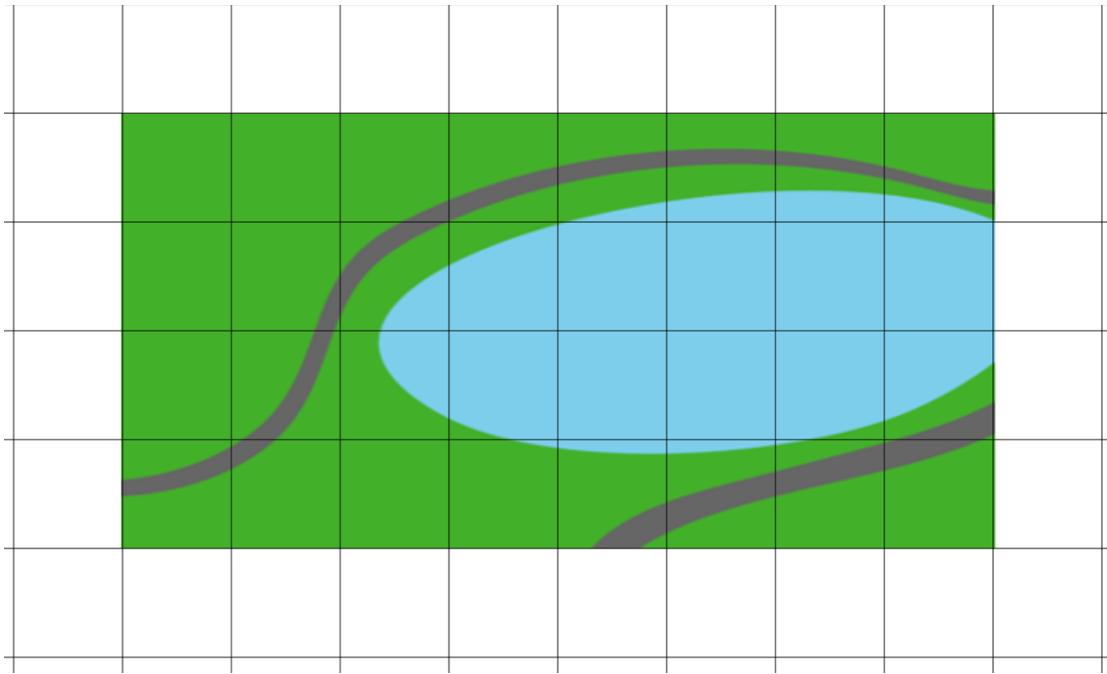
1. What are some features you notice about the map?

2. How would the scale help you if you were planning a trip?

3. How could the gridlines help you determine distance?

5.3.2 Guided Instruction: Scale Factor as a Ratio

Below is a scale drawing of a park where the scale is 1 centimeter (cm) to 90 meters (m). Each gridline represents 1 cm.



1. Fill in the table with the appropriate dimensions.

	Length	Width	Perimeter
Park Drawing	8 cm		24 cm
Actual Park		360 m	

2. Complete the ratio comparing the lengths from the table. Simplify if possible.

$$\frac{\boxed{} \text{ m}}{8 \text{ cm}}$$

3. Complete the ratio comparing the widths from the table. Simplify if possible.

$$\frac{360 \text{ m}}{\boxed{} \text{ cm}}$$

4. Write the perimeters as a ratio. Simplify if possible.

$$\frac{\boxed{} \text{ m}}{24 \text{ cm}}$$

5. What do you notice about the three ratios and the scale factor?
6. Complete the statement.

The scale or scale factor can be written in the form of a fraction. The ratios of the

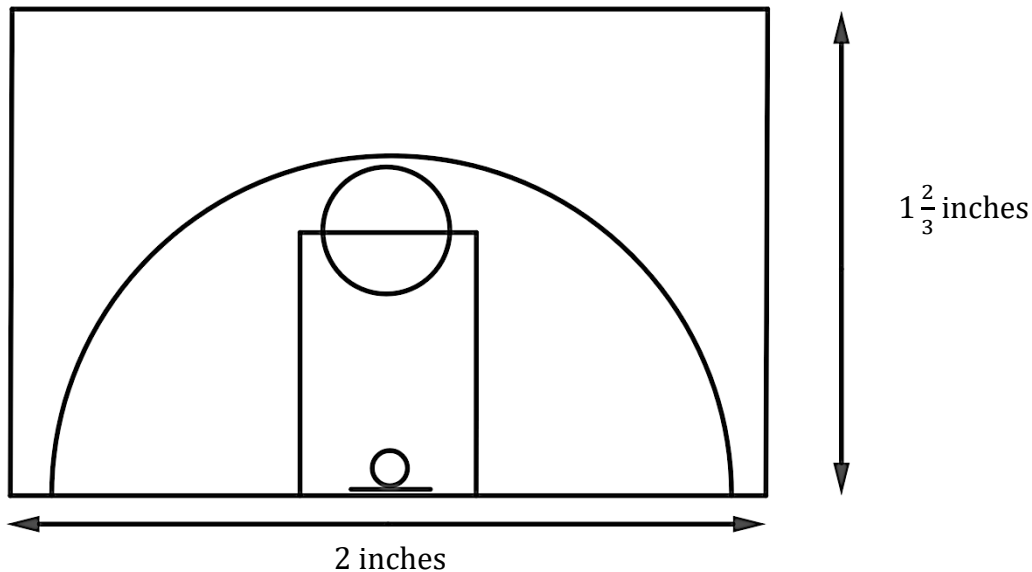
corresponding sides are ☐ equivalent
☐ not equivalent to the scale factor.

7. Santino tells Sarai that she could have found the perimeter using just the scale factor and the perimeter of the drawing. Explain how Santino could find the perimeter of the actual park using only the scale factor and the perimeter of the drawing.

5.3.3 Collaboration: Scales, Perimeter, and Area

Work with your partner to complete the following.

1. Vincent proposes an idea to the City Council to install a basketball hoop along with a half-court at his local park. His design is a scale drawing with length of 2 inches (in.) and width of $1\frac{2}{3}$ in.



- a. What is the actual length and width in feet (ft) of Vincent's planned basketball court if the scale is $\frac{20 \text{ ft}}{1 \text{ in.}}$?
- b. The council tells Vincent his plans will be approved if the court fits on a lot that measures 50 ft long and 25 ft wide. Explain whether Vincent's current plans for the court will fit.

- c. Which scale below will allow Vincent to fall under the parameters given by the council while creating the biggest court? Explain your selection.

☐ $\frac{18 \text{ ft}}{3 \text{ in.}}$

☐ $\frac{15 \text{ ft}}{1 \text{ in.}}$

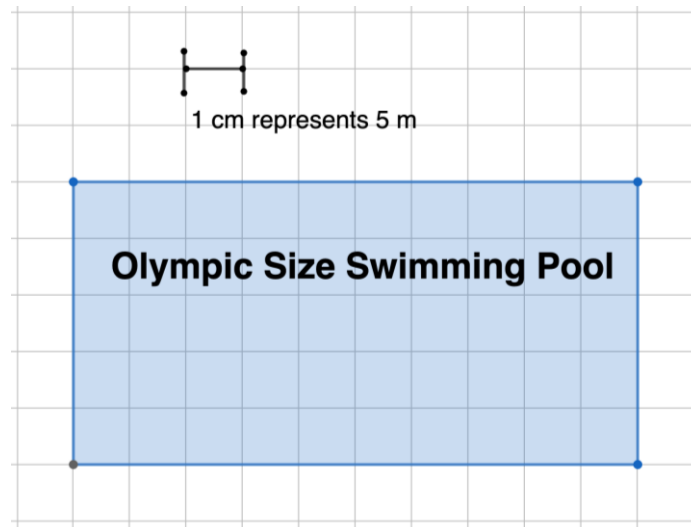
- d. After reviewing Vincent's work, the council agreed that using a scale of 15 feet (ft) to 1 in. would be acceptable.

The council will use the area of the court to calculate the cost of materials in square feet.

Use the scale $\frac{15 \text{ ft}}{1 \text{ in.}}$ and the drawing to determine the area of the actual basketball court.

5.3.4 Guided Practice: Real-World Problems Involving Area and Perimeter

1. A scale drawing of an Olympic-size swimming pool is shown on the grid where each 1 centimeter (cm) gridline represents 5 meters (m).

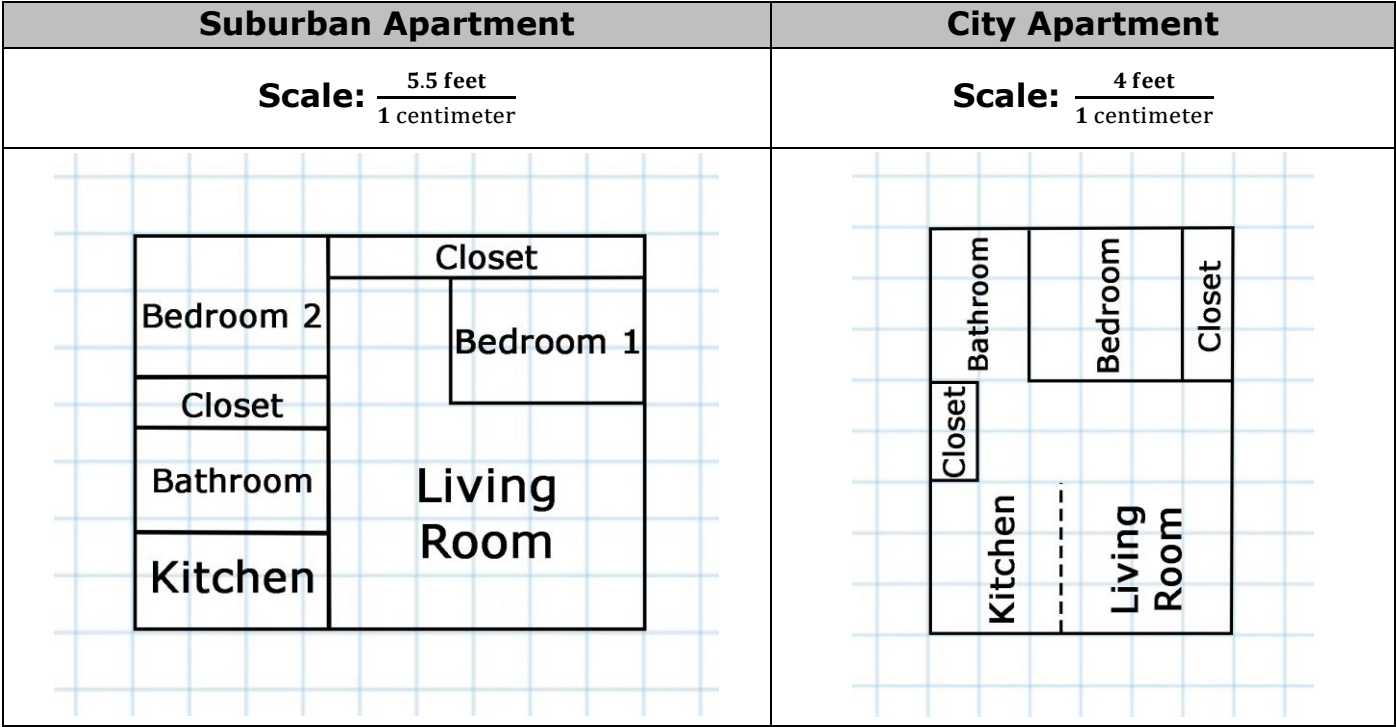


- a. Determine the actual dimensions of an Olympic-size swimming pool in meters (m).

Shelby wanted to build an Olympic-size swimming pool but did not have the space. So instead, she will build a scaled-down model.

- b. What is the perimeter of the pool in the scale drawing?
- c. What is the perimeter of the actual pool if the scale factor of the drawing was changed to $\frac{1 \text{ m}}{2 \text{ cm}}$?
- d. Calculate the perimeter of the pool if the drawing scale was changed to $\frac{2 \text{ m}}{5 \text{ cm}}$.

2. The models of two apartments are shown with the scales used to create them. Each gridline represents 1 centimeter (cm).



- a. Complete the tables to find the dimensions and areas of both apartments.

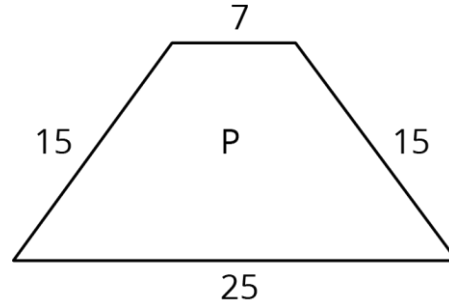
Suburban Apartment			
Scale	Length	Width	Area
1	9 cm		cm ²
$\frac{5.5 \text{ ft}}{1 \text{ cm}}$			ft ²

City Apartment			
Scale	Length	Width	Area
1		8 cm	cm ²
$\frac{4 \text{ ft}}{1 \text{ cm}}$			ft ²

- b. Which apartment is larger?
- c. Immanuel is looking for an apartment with as much closet space as possible. Explain which apartment you would recommend for him.

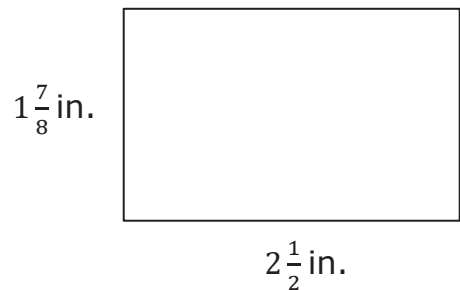
5.3.5 Your Turn!

1. Quadrilateral P is shown, where all measurements are in centimeters (cm).



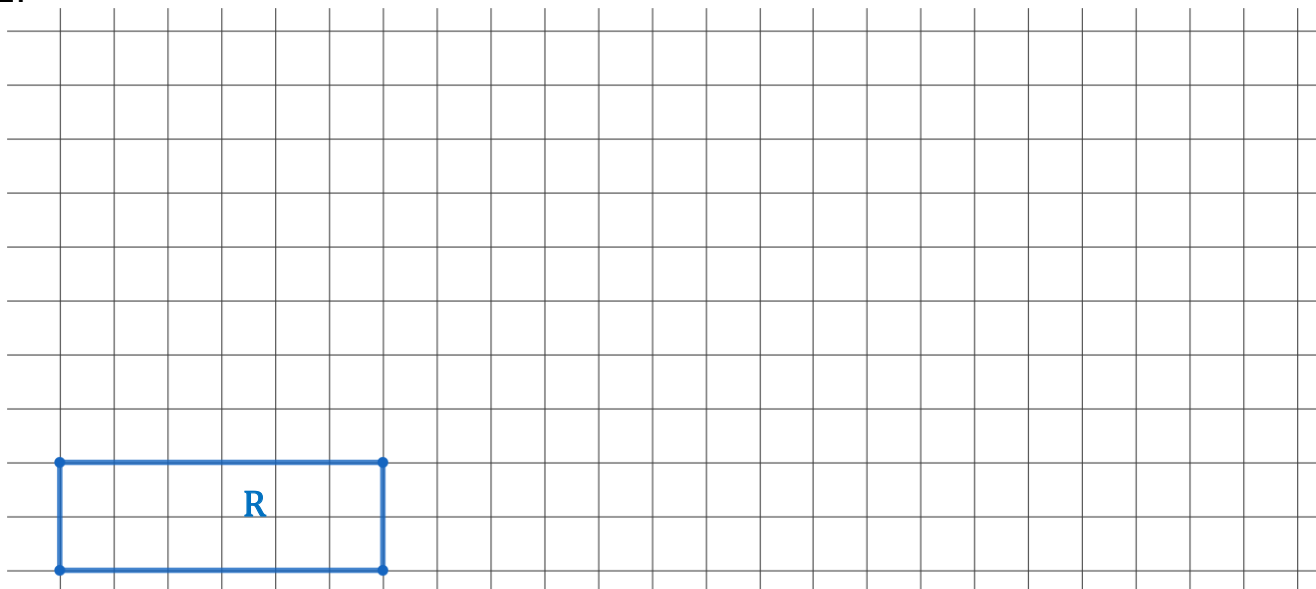
Quadrilateral R is a scaled copy of Quadrilateral P created using a scale factor of $\frac{5}{2}$. What is the perimeter of Quadrilateral R?

2. A scale drawing of the floor plan of Mariko's workshop is shown. If the scale is $\frac{40 \text{ ft}}{1 \text{ in.}}$, find the area, in square feet, of Mariko's actual workshop.



5.3.6 Wrap-Up: Ticket Out the Door

1. Polygon R has a length of 6 units and a width of 2 units.
- 2.



- a. On the grid, draw a scaled copy of Polygon R using a scale factor of 3.
- b. Explain how the perimeter of the new polygon compares to that of Polygon R .
- c. Explain how the area of the new polygon compares to that of Polygon R .